

No Mixed Partial Implies Additive Separation

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Abstract

A brief note to complete an observation, near the end of lecture on Nov. 17, on certain simple partial differential equations.

1 Solving $\frac{\partial^2 u}{\partial x \partial y} = 0$

Perhaps one of the simplest partial differential equations (PDEs) to start studying is $\frac{\partial^2 u}{\partial x \partial y} = 0$. Let us prove the following result two different ways. Recall that *univariate* simply means “depending on just one variable”.

Theorem 1.1 *Suppose u is a differentiable function defined on the open square $(-1, 1) \times (-1, 1)$ satisfying $\frac{\partial^2 u}{\partial x \partial y} = 0$. Then $u(x, y) = F(x) + g(y)$ for some (univariate) functions F and g differentiable on $(-1, 1)$.*

Remark 1.2 *We can in fact be much more general: we can replace $(-1, 1)$ and $(-1, 1) \times (-1, 1)$ respectively by U and $U \times U$, where U is any open subset of $\mathbb{C}$. We used the interval $(-1, 1)$ just to start simply. $\diamond$*

I started a proof at the end of lecture on November 17, 2011, so for the sake of completeness I give it in full now. Afterward, I’ll give a second proof. Feel free to prefer either proof (but at least prefer one!). Please also bear in mind that while I won’t be asking you to prove any theoretical results on the final, being nimble with theory helps guarantee a true understanding of the material.

To prepare for the first proof, let us observe a result that I didn’t have time to mention on Nov. 17.

Proposition 1.3 *Suppose u is a function of two variables x and y , continuous on an open rectangle $(a, b) \times (a, b)$, and g is a function differentiable on (a, b) . Then $\frac{\partial u}{\partial y} \Big|_{(x,y)=(x_0,y_0)} = g(y_0)$ for all $(x_0, y_0) \in (a, b) \times (a, b)$ implies that $u(x, y) = f(x) + G(y)$ on $(a, b) \times (a, b)$ for some univariate functions f and G , differentiable on (a, b) , with $G' = g$. ■*

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Put another way, the proposition tells us that, in the plane, integrating a function in y with respect to y results in a new function in y , *plus* a new function in x . The preceding proposition is actually quite easy to prove: First, the Fundamental Theorem of Calculus tells you that there must be a univariate function G with $u = G$ satisfying $\left. \frac{\partial u}{\partial y} \right|_{(x,y)=(x_0,y_0)} = g(y_0)$ for all $(x_0, y_0) \in (a, b) \times (a, b)$. Then, applying the limit definition of derivative, you immediately find (by holding x constant) that any other u satisfying the hypotheses differs from G by a function depending only on x .

Proof #1 for Theorem 1.1: First, we simply integrate $\frac{\partial^2 u}{\partial x \partial y}$ with respect to x . This is accomplished by applying Proposition 1.3 (with the variables reversed, and g the 0 function) to obtain that $\left. \frac{\partial u}{\partial y} \right|_{(x,y)=(x_0,y_0)} = g(y_0)$ for some function g differentiable on $(-1, 1)$. Applying Proposition 1.3 one last time, we are done. ■

Assuming one is comfortable with infinite series, our second proof is arguably more elementary.

Proof #2 for Theorem 1.1: Since $\frac{\partial^2 u}{\partial x \partial y}$ is identically 0 on $(-1, 1) \times (-1, 1)$, this implies that $\frac{\partial^2 u}{\partial x \partial y}$ is infinitely differentiable (since the 0 function is). As such, we can then consider the bivariate Taylor expansion $u(x, y) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} c_{i,j} x^i y^j$ and simply solve for the coefficients.

In particular, assuming uniform convergence, our differential equation implies that

$$\frac{\partial^2}{\partial x \partial y} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} c_{i,j} x^i y^j = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{\partial^2}{\partial x \partial y} c_{i,j} x^i y^j = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} c_{i,j} i j x^{i-1} y^{j-1}$$

is identically zero. This in turn implies that *all* the coefficients $c_{i,j}$ with $i, j \geq 1$ are zero. So the only coefficients $c_{i,j}$ left are those with $i = 0$ or $j = 0$. In other words, we can write $u(x, y) = c_{0,0} + \left(\sum_{i=1}^{\infty} c_{i,0} x^i \right) + \left(\sum_{j=1}^{\infty} c_{0,j} y^j \right)$. So we are done. ■

References

[Rud76] Rudin, Walter, *Principles of Mathematical Analysis*, 3rd edition, McGraw-Hill, 1976.